

Re-processing Perturb-Seq datasets from FASTQ

First Results for the Unified Pipeline

Kirill Tsukanov

Senior Full Stack Developer & Data Engineer

`ktsukanov@ebi.ac.uk`

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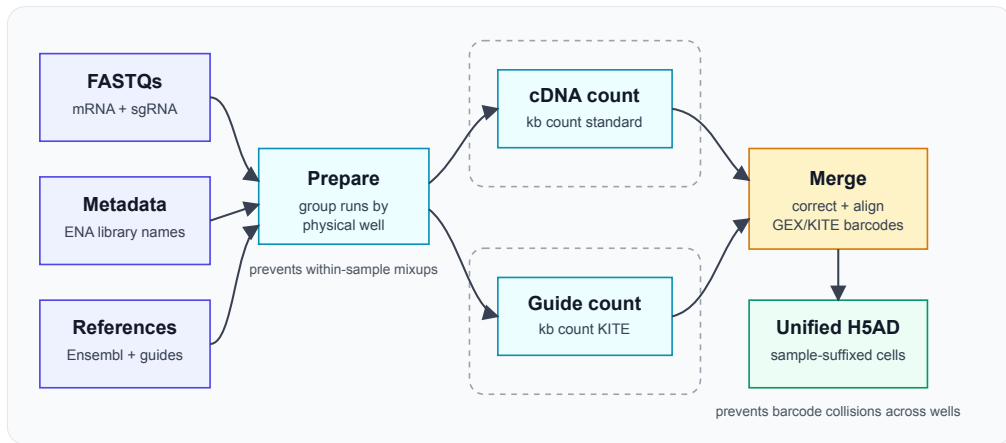
Why re-process from FASTQ?

- ▶ **Consistency:** scPerturb datasets are usually deposited as processed H5AD files, but they come from different pipelines, references, and filtering choices.
- ▶ **Comparable outputs:** Re-processing gives us a common count matrix layout, common metadata expectations, and a consistent way to attach guide counts.
- ▶ **Quality control:** Raw reads let us apply the same checks across studies instead of inheriting every author-specific decision.
- ▶ **Catalogue expectation:** Users expect harmonised data in the Perturbation Catalogue, not only links to heterogeneous processed files.

Why Kallisto + Bustools?

- ▶ **Speed:** Kallisto + Bustools is much faster than Cell Ranger while giving very comparable expression quantification.
- ▶ **Resource fit:** We have adequate, but not unlimited, compute and storage. The pipeline must work within those limits.
- ▶ **Full catalogue scale:** We want to process all datasets we have, and be able to reprocess them if the pipeline improves.
- ▶ **Future-proofing:** New perturb-seq datasets are expected to be very large, so the pipeline needs to scale well rather than depend on throwing more hardware at each run.

Pipeline Architecture



Development Issues Solved

- ▶ **ENA FASTQs missing technical reads:** Historical ENA FASTQs for this dataset do not include the barcode/UMI reads. Current workaround: SRA download with technical reads included. We notified ENA; they are aware and plan to fix it, so future runs may use ENA directly.
- ▶ **Sample grouping:** SRRs and lanes must be grouped by physical well before counting. This is currently parsed from ENA library names; later versions may autodetect this.
- ▶ **Barcode collisions:** Final concatenation adds sample suffixes so identical 10x barcodes from different wells remain distinct.
- ▶ **GEX/KITE matching:** Guide barcodes are not exactly the same as expression barcodes in this dataset. The pipeline applies the required KITE barcode correction before alignment.

Pipeline Execution

```
1 nextflow run main.nf \  
2   -profile slurm,singularity \  
3   --fastq_dir $HPS_PATH/perturb_seq_fastq/SAMN40972597 \  
4   --metadata_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/ena_metadata.tsv \  
5   --outdir $HPS_PATH/perturb_seq_fastq/results \  
6   --chemistry 10xv3 \  
7   --transcriptome_fa $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.dna.  
    primary_assembly.fa.gz \  
8   --gtf $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.111.gtf.gz \  
9   --features_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/features.tsv
```

- ▶ The sample sheet is generated automatically from ENA metadata.
- ▶ mRNA and sgRNA workflows run in parallel per sample.
- ▶ Guide counts are merged from unfiltered KITE output, then aligned to filtered mRNA cells.

First Dataset: Nadig 2025 Jurkat

- ▶ The unified FASTQ-to-H5AD pipeline now runs end-to-end on the EBI Slurm cluster.
- ▶ The first full dataset has been processed: Nadig 2025 Jurkat.
- ▶ The comparison was run against the author-provided H5AD.
- ▶ Analysis compares shared cells and shared genes after barcode and gene-name alignment.
- ▶ The main question: does the reprocessed dataset reasonably reproduce the expression and guide signals?

First Dataset: Scale and Runtime

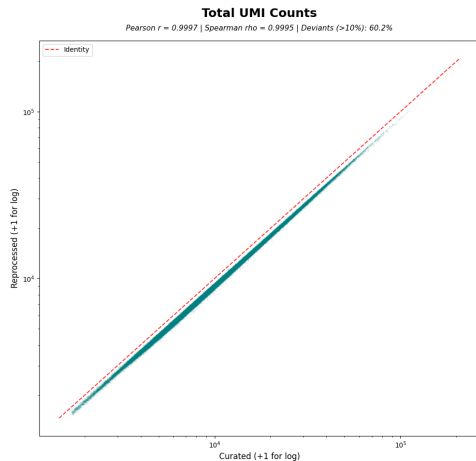
Input scale

- ▶ Full raw-data project: 2.618 TB and 2,688 reported FASTQ files.
- ▶ Jurkat subset processed here: 1,792 SRR runs, reported as 1,792 FASTQ files.
- ▶ Split by modality: 896 mRNA runs and 896 sgRNA runs.

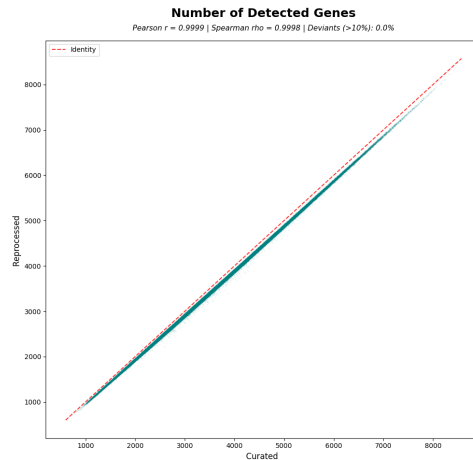
Pipeline run

- ▶ Wall time: **1h 18m**.
- ▶ Total compute: **504 CPU-hours**.
- ▶ Near-terabyte input processed with modest total compute.
- ▶ This is efficient enough to make catalogue-scale reprocessing realistic.

Cell-Level Expression Agreement

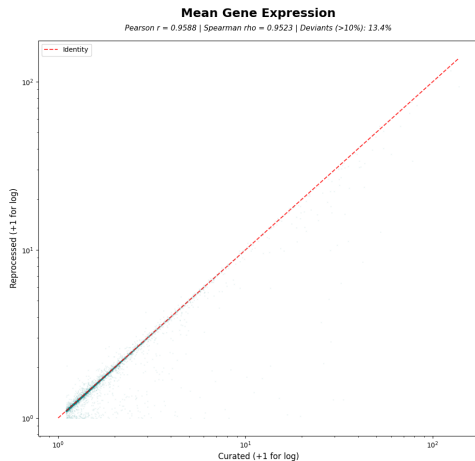


Correlation of total UMI counts per cell. Kallisto+Bustools (Reprocessed) generally identifies more UMIs per cell, especially in the high-sensitivity regime, likely due to transcript-level mapping vs gene-level quantification.

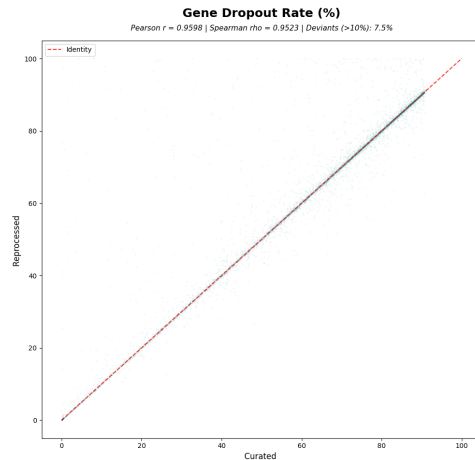


Correlation of unique genes detected per cell. The reprocessed pipeline recovers significantly more unique transcripts in a subset of cells, indicating higher detection sensitivity.

Gene-Level Expression Agreement



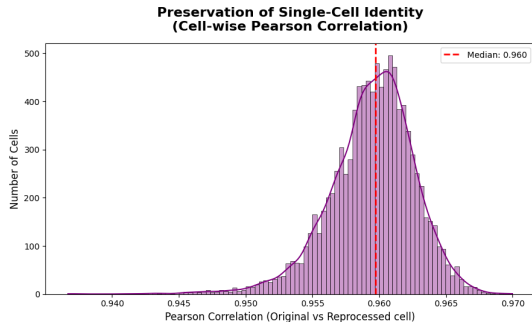
Average expression level for each shared gene. High correlation confirms that the biological signal is preserved across quantification methods.



Percentage of cells where a gene has zero counts. Lower values in Reprocessed indicate higher sensitivity for those genes.

Expression Vectors Are Preserved

- ▶ For each shared cell, we compare the original and reprocessed expression vectors over the same genes.
- ▶ The histogram shows the distribution of Pearson correlation values for a sample of 10,000 cells.



This distribution shows the correlation of the full expression profile for each individual cell against its exact counterpart in the reprocessed dataset. A strong peak near 1.0 confirms that single-cell identities are highly preserved.

Guide Recovery: Mixture Model Approach

- ▶ We use a **Gaussian-Poisson mixture model** to separate guide signal from background.
- ▶ The background is modeled as a Poisson distribution, while the true signal follows a Gaussian distribution.
- ▶ The decision threshold is automatically set at the crossover point where the probability of signal becomes higher than background.
- ▶ Cells are assigned a guide if at least one probe targets the gene and passes this threshold.
- ▶ Result: **205,457 / 214,522 (95.8%)** of cells match the original study assignments.

What We Have Now

- ▶ A working, reproducible FASTQ-to-H5AD pipeline for perturb-seq datasets.
- ▶ Both expression and guide calling are solved with very good alignment to original data.
- ▶ A comparison workflow that checks cell-level expression, gene-level expression, dropout, and perturbation labels.
- ▶ Pipeline is ready to scale to other datasets in the catalogue.

Next Steps

- ▶ **Parameter fine-tuning:** Optimizing the mixture model components for better signal separation.
- ▶ **Scale-out:** Start running the pipeline for all other available perturb-seq datasets.
- ▶ **Downstream analysis:** Run distance metrics between genes, plus the existing DEA and GSEA workflows.
- ▶ **Feedback:** This meeting is the last opportunity for feedback before we begin full-scale processing.